APPENDIX C

Second Quantization

C.1 Creation and Annihilation Operators

The name "second quantization" is misleading. This formalism has nothing to do with the further quantizing of quantum mechanics. It is just an alternative formulation of the usual quantum mechanics, which has turned out to be very useful for handling the many-body problem. It can be used for bosons and fermions, and we will give a short introduction and some important formulae.

We start with a complete orthogonal set of single-particle states $|\nu\rangle$, where ν stands for a set of quantum numbers, for example,

(i) space coordinate $\mathbf{r}$, spin $s \equiv s_z$ $\quad |\mathbf{r}, s\rangle$;
(ii) the quantum numbers of an oscillator basis $\quad |nljm\rangle$.

Orthogonality and completeness are expressed as

$$\langle \nu | \nu' \rangle = \delta_{\nu\nu'}, \qquad \sum_{\nu} |\nu\rangle\langle\nu| = 1. \tag{C.1}$$

(For continuous quantum numbers such as $\mathbf{r}$, the $\delta_{\nu\nu'}$ will mean $\delta(\mathbf{r}-\mathbf{r}')$ and the sum $\sum_{\nu}$ is to be replaced by $\int d^3r$.)

The coordinate representation of the state $|\nu\rangle$ is given by

$$\varphi_{\nu}(1) = \varphi_{\nu}(\mathbf{r}_1, s_1) = \langle \mathbf{r}_1, s_1 | \nu \rangle. \tag{C.2}$$

Starting with this set of single-particle states, we can construct a complete orthogonal set of totally *symmetric* N-body wave functions

$$\Phi_{\nu_1 \ldots \nu_N}(1, \ldots, N) = \mathfrak{N} \sum_{P} P \{ \varphi_{\nu_1}(1) \cdots \varphi_{\nu_N}(N) \}, \tag{C.3}$$

where the sum runs over all permutations $(\nu_1 \ldots \nu_N)$ of the numbers $(1 \ldots N)$ and $\mathfrak{N}$ is a normalization constant.

Any arbitrary, totally *symmetric* N-body wave function can be represented in this basis:

$$\Psi(1,\ldots,N) = \sum_{\nu_1 \ldots \nu_N} c_{\nu_1 \ldots \nu_N} \Phi_{\nu_1 \ldots \nu_N}(1,\ldots,N). \tag{C.4}$$

We can also give to each of the single-particle states a number $(\nu = 1, 2, \ldots)$ (in the case of continuous quantum numbers, we must first introduce a finite box) and characterize the wave function $\Phi_{\nu_1 \ldots \nu_N}$ by the "occupation numbers" $\{n_\nu\}$, which tell how often a particular number ν is contained in the N numbers $(\nu_1, \ldots, \nu_N)$. Obviously, we have

$$\sum_\nu n_\nu = N \tag{C.5}$$

and

$$\begin{aligned} \Phi_{\{n_\nu\}}(1,\ldots,N) &= \Phi_{\nu_1 \ldots \nu_N}(1,\ldots,N) \\ &= \frac{1}{\sqrt{N!}} \frac{1}{\sqrt{n_1! n_2! \ldots}} \sum_P P\{\varphi_{\nu_1}(1) \ldots \varphi_{\nu_N}(N)\}. \end{aligned} \tag{C.6}$$

Such a state describes a boson system. In complete analogy, we can construct totally antisymmetric-basis wave functions

$$\Phi_{\{n_\nu\}}(1,\ldots,N) = \frac{1}{\sqrt{N!}} \sum_P \operatorname{sign}(P) P\{\varphi_{\nu_1}(1) \ldots \varphi_{\nu_N}(N)\}. \tag{C.7}$$

They are called *Slater determinants* and describe fermion systems. In this case the numbers n_ν only take the values 0 or 1, otherwise (C.7) would vanish identically.

We can now construct a Hilbert space which contains a vacuum (no particle) $|-\rangle$, all the one-particle states, all the symmetrized (or antisymmetrized) two-particle states, and so on...

$$\mathfrak{H} = \{\mathfrak{H}_0, \mathfrak{H}_1, \mathfrak{H}_2, \ldots\}. \tag{C.8}$$

The wave functions $\Phi_{\{n_\nu\}}$ correspond to basis states $|n_1, n_2, \ldots\rangle$ in this Hilbert space, which are characterized by the occupation numbers (occupation number representation), such that

$$\Phi_{\{n_\nu\}}(1,\ldots,N) = \langle 1,\ldots,N | n_1, n_2 \ldots \rangle. \tag{C.9}$$

These states are orthonormalized

$$\langle n_1', n_2', \ldots n_\nu' \ldots | n_1, n_2 \ldots n_\nu \ldots \rangle = \delta_{n_1 n_1'} \cdot \delta_{n_2 n_2'} \cdots \delta_{n_\nu n_\nu'} \cdots. \tag{C.10}$$

First we shall study boson systems and define an "*annihilation operator*" B_ν by

$$B_\nu | n_1, n_2, \ldots, n_\nu, \ldots \rangle = \sqrt{n_\nu}\, | n_1, n_2, \ldots, n_\nu - 1, \ldots \rangle. \tag{C.11}$$

The operator B_ν lowers the occupation number in the state with the number ν by one. An N-body state goes over into an $(N-1)$-body state.

The matrix elements of B_ν are

$$\langle n_1', n_2', \ldots, n_\nu', \ldots | B_\nu | n_1, n_2, \ldots, n_\nu, \ldots \rangle = \sqrt{n_\nu}\, \delta_{n_1 n_1'} \ldots \delta_{n_\nu' n_\nu - 1} \cdots \quad \text{(C.12)}$$

or

$$\langle n_1, n_2, \ldots, n_\nu, \ldots | B_\nu^+ | n_1', n_2', \ldots, n_\nu', \ldots \rangle = \sqrt{n_\nu + 1}\, \delta_{n_1 n_1'} \ldots \delta_{n_\nu n_\nu' + 1} \cdots \quad \text{(C.13)}$$

This is valid for all basis states $\langle n_1, n_2 \ldots |$. We therefore find

$$B_\nu^+ | n_1, n_2, \ldots, n_\nu, \ldots \rangle = \sqrt{n_\nu + 1}\, | n_1, n_2, \ldots, n_\nu + 1, \ldots \rangle. \quad \text{(C.14)}$$

B_ν^+ "creates" a particle in the state with the number ν. Therefore, it is called a "*creation operator* in the state ν." It is the Hermitian conjugate operator to B_ν.

From this definition we gain the fact that

$$(B_\mu B_\nu^+ - B_\nu^+ B_\mu) | n_1, n_2, \ldots, n_\nu, \ldots, n_\mu \ldots \rangle$$

$$= \begin{cases} \left(\sqrt{n_\nu + 1}\, \sqrt{n_\mu} - \sqrt{n_\mu}\, \sqrt{n_\nu + 1}\right) | n_1, \ldots, n_\nu + 1, \ldots, n_\mu - 1, \ldots \rangle = 0 & \text{for } \nu \neq \mu, \\ \left(\sqrt{n_\nu + 1}\, \sqrt{n_\nu + 1} - \sqrt{n_\nu}\, \sqrt{n_\nu}\right) | n_1, \ldots, n_\nu, \ldots \rangle & \text{for } \nu = \mu, \end{cases} \quad \text{(C.15)}$$

and hence get the commutation relations.

$$\left[B_\mu, B_\nu^+ \right] = B_\mu B_\nu^+ - B_\nu^+ B_\mu = \delta_{\mu\nu}. \quad \text{(C.16)}$$

In the same way, we may show that

$$\left[B_\mu, B_\nu \right] = \left[B_\nu^+, B_\mu^+ \right] = 0. \quad \text{(C.17)}$$

The state with the occupation numbers $|0, 0, 0 \ldots \rangle = | - \rangle$ is the *vacuum*. We thus have

$$B_\nu | - \rangle = 0 \qquad \text{for all } \nu \quad \text{(C.18)}$$

and

$$| n_1, n_2, \ldots, n_\nu, \ldots \rangle = \frac{1}{\sqrt{n_1! n_2! \cdots n_\nu! \cdots}} \prod_\mu (B_\mu^+)^{n_\mu} | - \rangle. \quad \text{(C.19)}$$

The relation (C.11) follows from (C.16) to (C.19), which was our definition of the operators B_ν. We can therefore also go in the opposite direction and start with a set of operators B_ν, B_ν^+ which obey boson commutation relations and construct the many-body Hilbert space from (C.17) and (C.18).

The operator $B_\nu^+ B_\nu$ is called the particle-number operator for the state ν:

$$B_\nu^+ B_\nu | n_1, \ldots, n_\nu, \ldots \rangle = n_\nu | n_1, \ldots, n_\nu, \ldots \rangle. \quad \text{(C.20)}$$

We now address ourselves to a fermion system. We shall use small Latin letters a_ν^+, a_ν for the creation and annihilation operators of fermions. Since n_ν can only have the values 0 and 1, we may define the action of the

operators as

$$a_\nu|n_1,\ldots,n_\nu=1,\ldots\rangle=|n_1,\ldots,n_\nu=0,\ldots\rangle, \qquad a_\nu|n_1,\ldots,n_\nu=0,\ldots\rangle=0, \tag{C.21}$$

from which we get

$$a_\nu^+|n_1,\ldots,n_\nu=0,\ldots\rangle=|n_1,\ldots,n_\nu=1,\ldots\rangle, \qquad a_\nu^+|n_1,\ldots,n_\nu=1,\ldots\rangle=0, \tag{C.22}$$

and

$$\begin{aligned}\left[a_\mu, a_\nu^+\right]_+ &:= \{a_\mu, a_\nu^+\} := a_\mu a_\nu^+ + a_\nu^+ a_\mu = \delta_{\mu\nu},\\ \left[a_\mu, a_\nu\right]_+ &= \left[a_\mu^+, a_\nu^+\right]_+ = 0.\end{aligned} \tag{C.23}$$

The vacuum is again given by $|-\rangle=|0,0,\ldots\rangle$ and we have

$$a_\nu|-\rangle=0 \qquad \text{for all } \nu, \tag{C.24}$$

hence

$$|n_1,\ldots,n_\nu,\ldots\rangle=\prod_\mu (a_\mu^+)^{n_\mu}|-\rangle=a_{\nu_1}^+\ldots a_{\nu_N}^+|-\rangle. \tag{C.25}$$

C.2 Field Operators in the Coordinate Space*

Using the single-particle wave functions $\varphi_\nu(\mathbf{r},s)$ in Eq. (C.2) we can define creation and annihilation operators $a^+(\mathbf{r},s), a(\mathbf{r},s)$, which depend on the coordinates $\mathbf{r}$ and s†:

$$a(\mathbf{r},s)=\sum_\nu \varphi_\nu(\mathbf{r},s)a_\nu; \qquad a^+(\mathbf{r},s)=\sum_\nu \varphi_\nu^*(\mathbf{r},s)a_\nu^+. \tag{C.26}$$

With Eq. (C.1) we can invert this relation,

$$a_\nu=\sum_s \int d^3r\, \varphi_\nu^*(\mathbf{r},s)a(\mathbf{r},s), \qquad a_\nu^+=\sum_s \int d^3r\, \varphi_\nu(\mathbf{r},s)a^+(\mathbf{r},s), \tag{C.27}$$

and gain the commutators

$$\left[a(\mathbf{r},s), a^+(\mathbf{r}',s')\right]_+=\sum_{\nu\nu'}\varphi_\nu(\mathbf{r},s)\varphi_{\nu'}^*(\mathbf{r}',s')\left[a_\nu, a_{\nu'}^+\right]_+=\delta_{ss'}\delta(\mathbf{r}-\mathbf{r}') \tag{C.28}$$

and

$$\left[a(\mathbf{r},s), a(\mathbf{r}',s')\right]_+=\left[a^+(\mathbf{r},s), a^+(\mathbf{r}',s')\right]_+=0. \tag{C.29}$$

We can express the many-body wave function (C.7) by

$$\Phi_{\{n_\nu\}}(1,\ldots,N)=\frac{1}{\sqrt{N!}}\langle-|a(N)\ldots a(1)|n_1,n_2,\ldots n_\nu\ldots\rangle \tag{C.30}$$

*In the following we investigate only the case of fermions. Analogous considerations apply for bosons.

†This definition agrees with the convention $\varphi_\nu(\mathbf{r})=\langle\mathbf{r}|\nu\rangle$, and according to the definition (5.18) for a unitary transformation of the operators a_ν^+, $\varphi_\nu(\mathbf{r})$ corresponds to $D_{\nu\nu}^*$.

and

$$|n_1, n_2, \ldots n_\nu \ldots\rangle = \int d1 \ldots dN \frac{1}{\sqrt{N!}} \Phi_{\{n_\nu\}}(1, \ldots, N) a^+(1) \ldots a^+(N) |-\rangle. \tag{C.31}$$

C.3 Representation of Operators

Starting from a vacuum $|-\rangle$, we have expressed all states in the many-body Hilbert space $\mathcal{H}$ by creation and annihilation operators a_ν^+, a_ν. The same will be done for operators in the following. We have to distinguish between one- and two-body operators.

A *one-body operator* as, for example, the kinetic energy or the total momentum of an N-particle system, is given as the sum of N operators $\hat{f}_i$ which always act on the coordinate of the particle i:

$$\hat{F} = \sum_{i=1}^{N} \hat{f}_i. \tag{C.32}$$

Its matrix elements in the $|\nu\rangle$ representation are

$$f_{\nu\nu'} = \langle \nu | \hat{f} | \nu' \rangle, \tag{C.33}$$

that is,

$$\hat{f}_i \varphi_\nu(i) = \sum_{\nu'} f_{\nu'\nu} \varphi_{\nu'}(i). \tag{C.34}$$

The representation of $\hat{F}$ in the operators a_ν^+, a_ν is given by

$$\hat{F} = \sum_{\nu\nu'} f_{\nu\nu'} a_\nu^+ a_{\nu'}. \tag{C.35}$$

To show this, we have to prove

$$\sum_i \hat{f}_i \Phi(1, \ldots, N) = \langle 1, \ldots, N | \sum_{\nu\nu'} f_{\nu\nu'} a_\nu^+ a_{\nu'} | \Phi \rangle. \tag{C.36}$$

On the l.h.s, from Eqs. (C.30), (C.26) and (C.34) up to a factor $1/\sqrt{N!}$ we gain

$$\begin{aligned}
&\sum_i \hat{f}_i \langle - | a(N) \ldots a(i) \ldots a(1) | \Phi \rangle \\
&\quad = \sum_i \sum_{\nu_1 \ldots \nu_N} \hat{f}_i \varphi_{\nu_N}(N) \ldots \varphi_{\nu_i}(i) \ldots \varphi_{\nu_1}(1) \langle - | a_{\nu_N} \ldots a_{\nu_1} | \Phi \rangle \\
&\quad = \sum_i \sum_\nu \sum_{\nu_1 \ldots \nu_N} f_{\nu\nu_i} \varphi_{\nu_N}(N) \ldots \varphi_\nu(i) \ldots \varphi_{\nu_1}(1) \langle - | a_{\nu_N} \ldots a_{\nu_1} | \Phi \rangle.
\end{aligned}$$

This is identical to the r.h.s.:

$$\begin{aligned}
&\sum_{\nu\nu'} \sum_{\nu_1 \ldots \nu_N} f_{\nu\nu'} \varphi_{\nu_N}(N) \ldots \varphi_{\nu_1}(1) \langle - | a_{\nu_N} \ldots a_{\nu_1} a_\nu^+ a_{\nu'} | \Phi \rangle \\
&\quad = \sum_i \sum_\nu \sum_{\nu_1 \ldots \nu_N} f_{\nu_i\nu} \varphi_{\nu_N}(N) \ldots \varphi_{\nu_1}(1) \langle - | a_{\nu_N} \ldots a_\nu \ldots a_{\nu_1} | \Phi \rangle.
\end{aligned}$$

We give next a few examples:

The kinetic energy

$$\hat{T}=\sum_i \hat{t}_i=\sum_i \frac{-\hbar^2}{2m}\Delta_i, \tag{C.37}$$

$$\hat{T}=\sum_{\nu\nu'}\sum_s \int d^3r\, \varphi_\nu^*(\mathbf{r},s)\frac{-\hbar^2}{2m}\Delta\varphi_{\nu'}(\mathbf{r},s)a_\nu^+ a_{\nu'}$$

$$=\sum_s \int d^3r\, a^+(\mathbf{r},s)\frac{-\hbar^2}{2m}\Delta a(\mathbf{r},s). \tag{C.38}$$

The single-particle density (see Appendix D)

$$\hat{\rho}(\mathbf{r})=\sum_{i=1}^{N}\delta(\mathbf{r}-\hat{\mathbf{r}}_i). \tag{C.39}$$

$\hat{\mathbf{r}}_i$ is the coordinate operator of the ith particle; $\mathbf{r}$ is a number.

$$\hat{\rho}(\mathbf{r})=\sum_{\nu\nu'}\sum_s\int d^3r'\,\varphi_\nu^*(\mathbf{r}',s)\delta(\mathbf{r}-\mathbf{r}')\varphi_{\nu'}(\mathbf{r}',s)a_\nu^+ a_{\nu'}=\sum_s a^+(\mathbf{r},s)a(\mathbf{r},s). \tag{C.40}$$

The particle number

$$\hat{N}=\sum_\nu a_\nu^+ a_\nu=\sum_\nu\sum_{ss'}\int d^3r\, d^3r'\,\varphi_\nu(\mathbf{r},s)\varphi_\nu^*(\mathbf{r}',s')a^+(\mathbf{r},s)a(\mathbf{r}',s')$$

$$=\int\hat{\rho}(\mathbf{r})\,d^3r. \tag{C.41}$$

In the most general case, $\hat{f}$ will be an integral operator (a "nonlocal" one-particle operator):

$$\hat{f}\,\varphi(\mathbf{r},s)=\sum_{s'}\int d^3r'\, f_{ss'}(\mathbf{r},\mathbf{r}')\varphi(\mathbf{r}',s'). \tag{C.42}$$

A *two-particle operator* as, for example, a two-body interaction, is given by a sum of operators v_{ij} which act on the coordinates of the particles i and j.

$$V=\sum_{i<j=1}^{N} v_{ij}. \tag{C.43}$$

In the most general case, v_{ij} will be an integral operator in two variables, with matrix elements

$$v_{\mu\nu\mu'\nu'}=\langle\mu\nu|v|\mu'\nu'\rangle=\int d1\,d2\,d3\,d4\,\varphi_\mu^*(1)\varphi_\nu^*(2)v(1,2,3,4)\varphi_{\mu'}(3)\varphi_{\nu'}(4). \tag{C.44}$$

In complete analogy to Eq. (C.35), we can show that V can be written as

$$V=\tfrac{1}{2}\sum_{\mu\nu\mu'\nu'} v_{\mu\nu\mu'\nu'}a_\mu^+ a_\nu^+ a_{\nu'}a_{\mu'}=\tfrac{1}{4}\sum_{\mu\nu\mu'\nu'}\bar{v}_{\mu\nu\mu'\nu'}a_\mu^+ a_\nu^+ a_{\nu'}a_{\mu'}, \tag{C.45}$$

with the antisymmetrized matrix element

$$\bar{v}_{\mu\nu\mu'\nu'}=\langle\mu\nu|v|\mu'\nu'\rangle-\langle\mu\nu|v|\nu'\mu'\rangle. \tag{C.46}$$

Very often we use local two-body interactions of the form (we neglect spin)

$$v_{ij} = v(\mathbf{r}_i, \mathbf{r}_j). \tag{C.47}$$

In this case, we can verify Eq. (C.45) immediately with Eq. (C.40):

$$\begin{aligned}\tfrac{1}{2}\sum_{\mu\nu\mu'\nu'} v_{\mu\nu\mu'\nu'} a_\mu^+ a_\nu^+ a_{\nu'} a_{\mu'} &= -\tfrac{1}{2}\int d^3r\, d^3r'\, a^+(\mathbf{r}) a^+(\mathbf{r}') v(\mathbf{r},\mathbf{r}') a(\mathbf{r}) a(\mathbf{r}') \\ &= \tfrac{1}{2}\Big(\int d^3r\, d^3r'\, v(\mathbf{r},\mathbf{r}')\hat{\rho}(\mathbf{r})\hat{\rho}(\mathbf{r}') - \int d^3r\, v(\mathbf{r},\mathbf{r})\hat{\rho}(\mathbf{r})\Big) \\ &= \tfrac{1}{2}\Big(\sum_{i\neq j} v(\mathbf{r}_i,\mathbf{r}_j) - \sum_i v(\mathbf{r}_i,\mathbf{r}_i)\Big) = \sum_{i<j=1}^{N} v(\mathbf{r}_i,\mathbf{r}_j).\end{aligned} \tag{C.48}$$

C.4 Wick's Theorem

In practical applications of the second quantization, Wick's theorem has turned out to be very useful. It is a rule which allows a very simple reordering of a set of N-operators a or a^+, which have the property that the commutator (in the case of bosons) or the anticommutator (in the case of fermions) of two arbitrarily chosen operators of this set is a number.

We first define the T-product (time ordered product) of a product of operators $a(t_2), a(t_3), a^+(t_1)\ldots$ to be the one where the field operators have been reordered in such a way that the time arguments are increasing from right to left (an odd permutation gives a minus sign):

$$T\{a(t_1)a^+(t_3)a(t_2)a^+(t_4)\} = -a(t_1)a(t_2)a^+(t_3)a^+(t_4) \qquad t_1 > t_2 > t_3 > t_4. \tag{C.49}$$

In a *normal ordered* product, the field operators are ordered in such a way that all creation operators are to the left of all annihilation operators* (again, an odd permutation gives a minus sign):

$$N\{a_\nu a_\mu a_\rho a_\sigma^+\} =: a_\nu a_\mu a_\rho a_\sigma^+ := -a_\sigma^+ a_\nu a_\mu a_\rho. \tag{C.50}$$

The *contraction* $\widehat{UV}$ of two field operators U and V is defined as

$$\widehat{UV} = T\{UV\} - N\{UV\}. \tag{C.51}$$

With these definitions, *Wick's Theorem* [Wi 50] can be stated as:

$$\begin{aligned}T\{UVW\ldots XYZ\} &= N\{UVW\ldots XYZ\} + N\{\widehat{UV}W\ldots XYZ\} \\ &\quad + \cdots + N\{UVW\ldots X\widehat{YZ}\} + N\{\widehat{UV}W\ldots\widehat{XY}Z\} \\ &\quad + \cdots + N\{\widehat{UVW\ldots XYZ}\} + \cdots.\end{aligned} \tag{C.52}$$

The time ordered product of field operators is therefore equal to their normal ordered product plus the normal ordered products with one contraction (in all possible ways), plus the normal ordered product with two

* Instead of $N\{\ldots\}$ we also often use double dots $:\ldots:$.

contractions, and so on. Care has to be taken in the removal of a contraction out of a normal ordered product, as this can give a minus sign:

$$N\{\widehat{UVXY}\} = \widehat{UV}N\{XY\}; N\{\widehat{UVXY}\} = -\widehat{UX}N\{VY\}. \quad \text{(C.53)}$$

(For the proof of Wick's theorem see, e.g., Thouless [Th 61b].)

Wick's theorem is especially useful for the calculation of ground state expectation values of time ordered products of field operators (e.g., the expectation value of particle operators with respect to the quasi-particle vacuum in Chap. 6). The result is equal to the r.h.s. of (C.52), where all operators have been contracted in all possible ways.

If there are time-independent field operators, as in (C.45), then the given order is to be defined as time ordered. The contraction of time-independent operators is especially simple; for example, we get:

$$\widehat{a_\mu a_\nu^+} = a_\mu a_\nu^+ - N\{a_\mu a_\nu^+\} = \delta_{\mu\nu}. \quad \text{(C.54)}$$